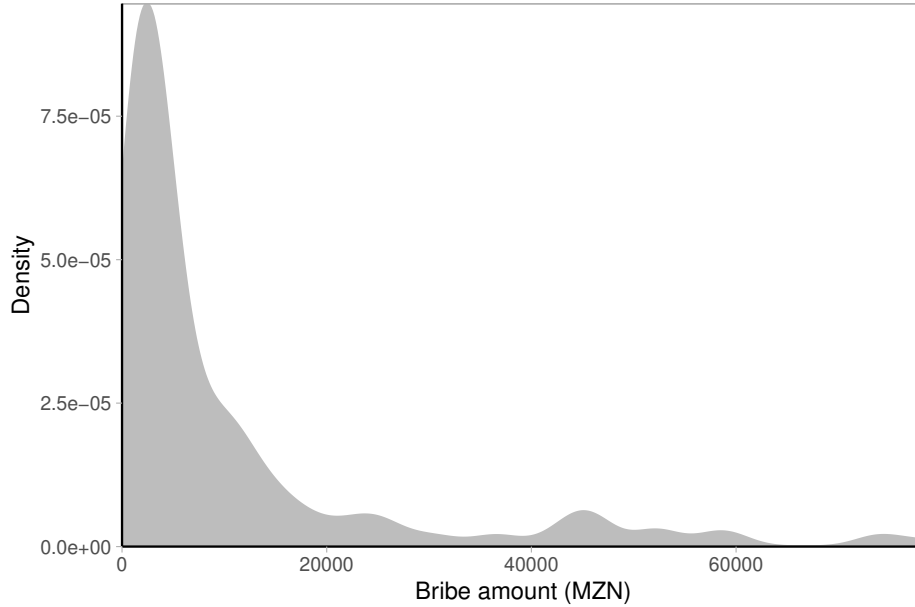


Figure 2: Density of bribe amount in Sequeira (2016)



Note: this figure shows a kernel density estimate of the bribe amount in Sequeira (2016), pooling across all observations with a positive bribe. The kernel density estimates are constructed using the default settings of the `stat_density` function in R.

energy-efficient stoves. They use the randomized price (p_i) as an instrument for whether an individual i buys an energy-efficient stove (D_i). They use this instrument to estimate the effects of stove-adoption on outcomes such as charcoal spending (Y_i).

Original specification and sensitivity to scale. Let X_i be a vector of control variables (including a constant). Berkouwer and Dean (2022) estimate

$$\text{arcsinh}(Y_i) = D_i\beta + X_i'\gamma + \epsilon_i \quad (11)$$

by two-stage least squares (TSLS), using p_i as an instrument for D_i .⁴⁰ (They also report results where spending is measured in levels.) The estimated coefficient $\hat{\beta}$ is an estimate of the LATE of stove adoption on the arcsinh of charcoal spending for instrument-compliers whose decision of whether to purchase the stove depends on the price offered in the experiment.⁴¹ In Berkouwer and Dean (2022), Y_i is measured as weekly charcoal spending in dollars. They obtain a coefficient of $\hat{\beta} = -0.50$ and write “[t]he 50 log point reduction corresponds to a 39 percent decrease in charcoal consumption [since $\exp(-0.50) = 1 - 0.39$]” (p. 3306).

⁴⁰More precisely, each observation i is an individual-by-week pair, and some (but not all) individuals are surveyed on multiple weeks. Standard errors are clustered at the respondent level.

⁴¹We use the phrase “instrument-compliers” to distinguish compliers for the instrument, whose value of $D(z)$ depends on z , from “compliers” discussed earlier who have $Y(1) > 0, Y(0) = 0$. Since the instrument takes on multiple values (i.e. multiple price offers), β corresponds to a weighted average of treatment effects across instrument-compliers for different values of the instrument (Angrist et al., 2000).

However, if we change the units of the outcome to annual charcoal spending in Kenyan shillings, the original currency in which charcoal spending was measured, the same specification yields an estimate of -0.44 . Relative to our previous applications, the change in the treatment effect estimates is fairly small for these choices of units, due to a small estimated extensive margin of 0.01 (see Table 1).⁴² Nevertheless, the fact that the treatment effects using an arcsinh-transformed outcome depend on the units should give us pause in interpreting them as percentages. Indeed, a percentage effect is not well-defined for someone who has non-zero spending under treatment and zero spending under the control, so an average individual-level percentage effect does not make sense if the treatment can affect whether one has any charcoal spending.

Berkouwer and Dean (2022) first discuss the LATE in levels, and then immediately afterwards state that the treatment effect for the arcsinh-transformed outcome “corresponds to a 39 percent decrease in charcoal consumption” (p. 3306). The main goal of taking the arcsinh transformation here thus appears to be to obtain a treatment effect with a percentage interpretation. We therefore next implement two approaches with an (approximate) percentage interpretation in this context.

Proportional LATE. One natural approach in this context is to estimate the proportional change in the average outcome for instrument-compliers, i.e. to estimate $\theta_{\text{ATE}\%}$ among the population of instrument-compliers. Put otherwise, we can express the LATE in levels as a percentage of the control mean for instrument-compliers. An estimate of the LATE in levels is naturally obtained using TSLS specification (11) with Y_i as the outcome, which yields an estimate of -2.46 . As described in Abadie (2002), we can likewise obtain an estimate of the control instrument-complier mean by using TSLS with $-(D_i - 1) \cdot Y_i$ as the outcome, which yields an estimate of 5.86. Putting these together, we obtain an estimate of $\theta_{\text{ATE}\%}$ for instrument-compliers of $-2.46/5.86 = -0.42$ ($\text{SE} = 0.046$), which suggests that average charcoal spending is 42% lower for instrument-compliers under treatment than under control.⁴³ If pollution is proportional to charcoal spending, then this parameter is economically relevant as it corresponds to the percentage reduction in pollution for instrument-compliers from gaining access to the efficient stove.

Lee bounds. Berkouwer and Dean (2022) benchmark their treatment effect estimates relative to engineering estimates of the efficiency gains of using an efficient stove relative to a non-efficient one. For this benchmarking exercise, it seems sensible to focus on the intensive-margin effect of the treatment—i.e., the treatment effect for instrument-compliers who would use a non-efficient stove if offered a high price and an efficient one if offered a low price. To do so, we can form Lee (2009)-type

⁴²We note, however, that the t -statistic for the effect on $\text{arcsinh}(Y_i)$ is rather sensitive here, changing from approximately 7 to 3 depending on the units.

⁴³The standard error was calculated via a non-parametric bootstrap with 1,000 draws, clustered at the respondent level. We note that with a binary instrument, an estimate of $\theta_{\text{Intensive}}$ for instrument-compliers can be obtained using Poisson IV regression (e.g. the `ivpoisson` command in Stata); see Angrist (2001). However, we are not aware of a LATE interpretation of Poisson IV regression with a multi-valued instrument, and thus do not pursue it here. Whether Poisson IV regression has such an interpretation with a continuous IV strikes us an interesting topic for future work.

bounds for the average treatment effect in logs for instrument-compliers who would have positive charcoal spending regardless of treatment status.⁴⁴

The bounds on $\theta_{\text{Intensive}}$ for instrument-compliers are $[-0.565, -0.538]$ (with SEs for the lower and upper bounds of 0.072 and 0.075).⁴⁵ This implies that for the instrument-compliers who would spend on charcoal regardless of treatment status, spending decreases by 54 to 56 log points. We note that the Lee bounds are fairly tight in this case, as tends to be the case when the extensive margin is small. It is also worth noting that in this example, the estimated treatment effects using $\text{arcsinh}(Y_i)$ —both in terms of weekly spending in dollars and in terms of annual spending in Kenyan shillings—fall outside of the Lee bounds, although they are fairly close to the upper bound when using weekly spending in dollars.

6 Conclusion

It is common in empirical work to estimate ATEs for transformations such as $\log(1+Y)$ or $\text{arcsinh}(Y)$ which are well-defined at zero and behave like $\log(Y)$ for large values of Y . We show that the ATEs for such transformations should not be interpreted as percentages, since they depend arbitrarily on the units of the outcome when there is an extensive margin. Further, we show that any parameter that is an average of individual-level treatment effects of the form $E_P[g(Y(1), Y(0))]$ must be scale-dependent if it is point-identified and well-defined at zero. We discuss several alternative approaches, including estimating scale-invariant normalized parameters (e.g. via Poisson regression), explicitly calibrating the value placed on the intensive versus extensive margins, and separately estimating effects for the intensive and extensive margins (e.g. using Lee bounds). We illustrate how these approaches can be applied in practice in three empirical applications.

⁴⁴The validity of the [Lee \(2009\)](#)-type bounds requires the “monotonicity” assumption that all instrument-compliers who would have some charcoal consumption when not buying an efficient stove would also have some charcoal consumption when buying an efficient stove, which seems reasonable. Note that this is a distinct assumption from the instrument monotonicity assumption needed for a LATE interpretation for instrumental variables ([Imbens and Angrist, 1994](#)), which in this context states that anyone who would buy a stove at a higher price would also buy at a lower price.

⁴⁵We obtain these estimates using the procedure in [Abadie \(2002\)](#), as described in detail in [Appendix E](#).

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